

UNIVERSITY ENTRANCE TEST FOR ADMISSION IN UNDER-GRADUATE PROGRAMMES (2018)

SERIES

A

Booklet No:

608377

Time allowed :

2 hours 15 minutes for PCB & PCM and 3 hours for PCBM

This booklet contains 180 questions and 25 pages
Do not open this booklet until you are asked by the invigilator

Instructions to be followed:

1. Immediately after opening the Question Booklet, ensure that it contains the number of pages and questions as depicted above. In case of any discrepancy, get the question booklet replaced.
2. There are 180 MCQs under Physics, Chemistry, Biology and Mathematics with 45 questions in each section. Students appearing under test stream PCB will attempt questions numbered between 1 to 135, those appearing under test stream PCM will attempt questions numbered between 1 to 90 and 136 to 180 and those appearing under test stream PCBM will attempt all the 180 questions. However, Mathematics section, separate OMR shall be provided after submission of filled OMR of PCB by the student to the invigilation staff.
3. Use only blue/black pen for writing your particulars like Name, Roll No., Series and Question Booklet No. on the OMR answer sheet at spaces provided. OMR answer sheets without mentioning Series or Question Booklet No. will not be evaluated.
4. Use only blue/black pen for writing your responses on the OMR answer sheet.
5. Do not write anything else on OMR answer sheet. For rough work use the spaces provided on question booklet. Erasing, cutting or overwriting on the OMR sheet in answering circles will be deemed as incorrect answer and shall result in zero marks.
6. The candidates should on demand show their admit cards to invigilators or any other authorized representative of SKUAST-K.
7. No candidate will be allowed to leave the examination hall before half time except under special circumstances and permission by Centre Superintendent.
8. The candidates shall hand over the OMR answer sheet/s to invigilator before leaving the examination hall.
9. Use of electronic gadgets such as calculators, mobiles, pagers etc are not allowed in the examination hall.
10. Any act of indiscipline/misconduct/use of unfair means shall result in forfeiture of candidate's right to sit in the examination. The decision of superintendent of the Centre in this regard shall be the final.
11. The cases pertaining to the use of unfair means will be governed by the rules and regulations of SKUAST-K.

Physics

- Q 1 The scientific principle involved in computers is :
 A EMI ~~Deleted~~ B Thermodynamics
 C Super conductivity D Amplification by population inversion
- Q 2 Dimensions of intensity are :
 A $L^0 M^1 T^{-3}$ B $L^1 M^1 T^{-2}$
 C $L^2 M^1 T^{-2}$ D $L^2 M^2 T^{-3}$
- Q 3 Length cannot be measured in :
 A Fermi B Micron
 C Debye D Light year
- Q 4 In an elevator moving vertically up with an acceleration 'g' the force exerted on the floor by a passenger of mass 'M' is :
 A Mg B $\frac{1}{2} Mg$
 C Zero D $2 Mg$
- Q 5 A bomb of mass 16 kg at rest explodes into two pieces of masses 4 kg and 12 kg. The velocity of the 12 kg mass is 4m/sec. The K.E. of the other mass is :
 A 288 J B 192 J
 C 96 J D 144 J
- Q 6 When a wheel is rolling on a level road, the direction of frictional force between the wheel and road is in :
 A Backward direction B Forward direction
 C Depends on speed D None of these
- Q 7 When sand is poured on a rotating disc, its angular velocity will :
 A Decrease B Increase
 C Remain constant D None of the above
- Q 8 The time period of a second's pendulum in a satellite is :
 A Zero B 2
 C Infinity D Depends on mass of body
- Q 9 A long spring is stretched by 2 cm and its P.E. is V. If the spring is stretched by 10 cm, its P.E. will be :
 A $\frac{V}{5}$ B $\frac{V}{25}$
 C 5V D 25V

- Q 10 At which of the following temperatures, the value of surface tension of water is minimum?
 A 4°C B 25°C
 C 50°C D 75°C
- Q 11 Bulk modulus of water is $2 \times 10^9 \text{ N/m}^2$. The pressure required to increase the density of water by 0.1 % in N/m^2 is:
 A 2×10^8 B 2×10^9
 C 2×10^6 D 2×10^4
- Q 12 Let Q and W denote the amount of heat given to an ideal gas and the work done by it in an adiabatic process, then:
 A $Q = W$ B $Q = 0$
 C $W = 0$ D None of the above
- Q 13 The S I unit of specific heat is:
 A $\text{J Kg}^{-1} \text{K}^{-1}$ B $\text{J Kg}^{-1} \text{K}$
 C J Kg K^{-1} D $\text{J}^{-1} \text{Kg K}^{-1}$
- Q 14 A $4 \mu\text{F}$ capacitor is charged to 400 V and then the plates are joined through a resistor. Heat produced in resistor is:
 A 0.64 J B 0.32 J
 C 0.16 J D 1.28 J
- Q 15 If earth is supposed to be metallic sphere, its capacity will be nearly:
 A 700 PF B 700 F
 C $6.4 \times 10^6 \text{ F}$ D $711 \mu\text{F}$
- Q 16 Three resistances each of 4Ω connected to form a triangle. The resistance between any two terminals is:
 A 12Ω B 6Ω
 C $\frac{8}{3} \Omega$ D 2Ω
- Q 17 Identify the paramagnetic substance
 A Copper B Hydrogen
 C Aluminium D Nickel
- Q 18 Unit of self inductance is:
 A Volt amp^{-1} B Joule amp^{-1}
 C Volt amp D $\text{Volt second amp}^{-1}$

- Q 19 Maximum value of current from AC source of 50 Hz is 5 A. The time taken by current to grow from 0 to 2.5 A is :
 A 0.6 ms
 B 1.67 ms
 C 5 μ s
 D 0.5 μ s
- Q 20 A plano-convex lens has focal length $f = 20$ cm. If its plane surface is polished (silvered), the new focal length will be :
 A 10 cm
 B 15 cm
 C 20 cm
 D 25 cm
- Q 21 A man wants to see two poles separately situated at 11 km. The minimum approximate distance between the poles is :
 A 1 m
 B 3 m
 C 0.5 m
 D 5 m
- Q 22 To get three images of a single object, one should have two plane mirrors at an angle of :
 A 30°
 B 60°
 C 90°
 D 120°
- Q 23 The radius of Bohr's first orbit is a_0 . The electron in n th orbit has a radius :
 A $n a_0$
 B a_0/n
 C $n^2 a_0$
 D a_0/n^2
- Q 24 Which of the following logic gates is a universal logic gate ?
 A OR
 B AND
 C NOT
 D NAND
- Q 25 A simple pendulum hangs from the roof of train. The string is inclined towards the rear of the train. What is the nature of motion of the train ?
 A Uniform
 B Accelerated
 C Retarded
 D At rest
- Q 26 A wooden raft of mass 30 kg is floating on the surface of water. Density of raft is 600 Kg m^{-3} . How much mass can be placed on the raft to make it just sink?
 A 30 Kg
 B 40 Kg
 C 50 Kg
 D 20 Kg
- Q 27 The temperature of the source of a Carnot heat engine is 0°C and that of sink is -39°C . The efficiency of the heat engine is :
 A Zero
 B 14.3 %
 C 39 %
 D None of the above

Q 28 A one coulomb charge is placed on an insulated stand at the centre of a spherical conductor of radius 1 m. The sphere is given a charge of 1 C. The electrostatic force experienced by the charge at the centre will be :

- ☒ A Zero
☒ C 9×10^9 N

- B 1 N
D None of the above

Q 29 Which of the following can not produce colours with white light ?

- A Dispersion
C Diffraction

- B Interference
☒ D Polarization

Q 30 If the units of force and length are doubled, the unit of energy will be :

- ☒ A $\frac{1}{4}$ times
C 2 times

- B $\frac{1}{2}$ times
☒ D 4 times

Q 31 Maximum and minimum magnitudes of the resultant of two vectors of magnitudes P and Q are in the ratio 3 : 1, which of the following relation is true?

- ☒ A $P = 2Q$
C $PQ = 1$

- B $P = Q$
D None of the above



Q 32 What makes it difficult to separate two glass sheets having a drop of water between them ?

- ☒ A Viscosity
☒ C Surface tension

- B Gravity
D Atmospheric pressure

Q 33 A circle has been drawn round a point positive charge (+q) at its centre. The work done in taking a unit positive charge once round is :

- A 1 J
☒ C q J

- B $2\pi q$ J
☒ D Zero

Q 34 An electron and a proton enter a magnetic field with equal velocities, which one of them experiences more force ?

- A Electron
☒ C Both experience same force

- B Proton
☒ D It cannot be predicted

Q 35 Convex lens of power 4 D and a concave lens of power 3 D are placed in contact. What is the equivalent power of the combination ?

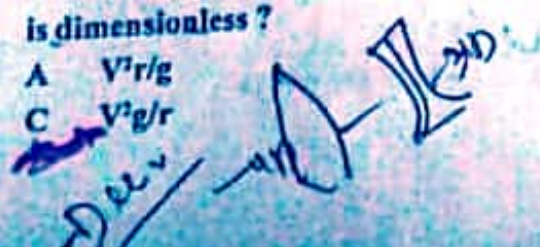
- A 7 D
☒ C 1 D

- B $\frac{4}{3}$ D
☒ D $\frac{3}{4}$ D

Q 36 Given that V is the speed, r is radius and g is acceleration due to gravity. Which of the following is dimensionless ?

- A $V^2 r / g$
☒ C $V^2 g / r$

- B V^2 / rg
D $V^2 rg$



V^2 / rg

- Q 37 To enable a particle describe a circular path. What should be the angle between its velocity and acceleration ?
A 0
☒ C 90°
B 45°
D 180°
- Q 38 In the elastic collision of a heavy vehicle moving with a velocity 10 ms^{-1} and small stone at rest, the stone will fly away with a velocity equal to :
A 5 ms^{-1}
☒ C 20 ms^{-1}
B 10 ms^{-1}
D 40 ms^{-1}
- Q 39 A water fall is 84 m high. Assuming that half of the kinetic energy of the falling water is converted to heat, what will be the rise in temperature ? Take $g = 10 \text{ ms}^{-2}$
☒ A 0.1° C
C 0.3° C
B 0.2° C
D 0.4° C
- Q 40 A $2 \mu\text{F}$ capacitor is charged to 100 V and then its plates are connected by a wire. How much heat will be produced ?
A 1 J
☒ C 0.01 J
B 0.1 J
☒ D 0.001 J
- Q 41 When two semiconductors of p and n type are brought into contact, they form a p-n junction, which acts like a :
A Conductor
C Amplifier
B Oscillator
☒ D Rectifier
- Q 42 If the earth stops rotating about its axis, the acceleration due to gravity will remain unchanged at:
A Equator
C Latitude 60°
B Latitude 45°
☒ D Poles
- Q 43 In streamline flow, the velocity of the liquid in contact with the containing vessel is :
☒ A Zero
C Large
B Minimum but not zero
D Infinite
- Q 44 In a transformer the voltage is stepped up from 220 V to 440 V . If the current in the primary was 1.0 ampere , that in the secondary should be :
☒ A 0.5 A
C 1 A
B Between 0.5 and 1 A
D More than 1 A
- Q 45 According to the Kirchhoff's law the sum of the products of current and resistance as well as emfs in a closed loop is :
A Greater than zero
C Less than zero
B Zero
D Determined by the emf

Chemistry

- Q 46 The dual nature of radiation is explained by the set of phenomena
 A Scintillation and interference ☒ B Photoelectric effect and diffraction
 C Scintillation and radioactivity ☒ D Photoelectric effect and scintillation
- Q 47 Which set of quantum numbers (n,l,m,s) is not possible for "f" orbital
 A 4, 2, 2, +1/2 ☒ B 4, 3, -2, -1/2
 C 5, 3, 0, +1/2 ☒ D 5, 3, 2, -1/2
- Q 48 Transition metals have the general electronic configuration
 A $ns^2 nd^{1-10}$ ☒ B $ns^2 np^1 (n-1) d^{1-10}$
 C $ns^{1-2} (n-1) d^{1-10}$ ☒ D $ns^2 np^6 (n-1) d^{1-10}$
- Q 49 In which of the following cases reaction is spontaneous at all temperatures
 A $\Delta H > 0; \Delta S > 0$ ☒ B $\Delta H < 0; \Delta S > 0$
 C $\Delta H < 0; \Delta S < 0$ ☒ D $\Delta H < 0; \Delta S = 0$
- Q 50 The solubility of a product of silver bromide is 3.3×10^{-13} , its solubility will be
 A 57.4×10^{-7} moles/litre ☒ B 0.574×10^{-7} moles/litre
 C 0.574×10^{-4} moles/litre ☒ D 5.74×10^{-7} moles/litre
- Q 51 The reagent commonly used to determine hardness of water titrimetrically is
 A Oxalic acid ☒ B Sodium citrate
 C Sodium thiosulphate ☒ D Disodium salt of EDTA
- Q 52 Borax is used as a cleansing agent because on dissolving in water it gives
 A Alkaline solution ☒ B Acidic solution ☒
 C Bleaching solution ☒ D Colloidal solution
- Q 53 Copper pyrites are concentrated by
 A Electromagnetic method ☒ B Gravity method
 C Froth floatation method ☒ D All of above
- Q 54 How many σ and π bonds are there in a molecule of tetracyanoethene
 A 9 σ , 7 π ☒ B 5 σ , 9 π
 C 5 σ , 8 π ☒ D 9 σ , 9 π
- Q 55 An organic compound with molecular formula C_6H_{12} upon ozonolysis gave only acetone. The compound is
 A 2,3-dimethyl-1-butene ☒ B 3-hexene
 C 2-hexene ☒ D 2,3-dimethyl-2-butene

- Q 56 $C_6H_6 + CH_3Cl \xrightarrow{\text{Anhydrous } AlCl_3} C_6H_5CH_3 + HCl$. The reaction is called :
 A Kolbe's reaction
 B Wurtz reaction
 C ☒ Wurtz Fittig reaction
 D ☒ Friedel Craft's reaction
- Q 57 Which is used in the formation of Nylon-66
 A Sulphur hexafluoride
 B ☒ Adipic acid
 C ☒ Pthalic acid
 D ☒ Sulphurous acid
- Q 58 The vitamin which is neither soluble in water nor fat is
 A ☒ Phyloquinone
 B ☒ Biotine
 C ☒ Ergocalciferol
 D Thiamine
- Q 59 The density of 2.05 M acetic acid in water is 1.02 g/cc. The molality of the solution will be
 A 1.03 m
 B 3.07 m
 C ☒ 2.29 m
 D 5.135 m
- Q 60 Gold numbers of protective colloids A, B, C and D are 0.50, 0.01, 0.10 and 0.005 respectively, the correct order of their protective power is
 A $B < D < A < C$
 B ☒ $D < A < C < B$
 C $C < B < D < A$
 D ☒ $A < C < B < D$
- Q 61 Standard entropy of X_2 , Y_2 and XY_2 are 60, 40 and 50 $JK^{-1} \text{ mole}^{-1}$ respectively. For the reaction $Y_2 + X_2 + 3/2 Y_2 \longrightarrow XY_2$, $\Delta H = -30 \text{ KJ}$, to be at equilibrium, the temperature will be
 A 1000 K ☒ Deleted
 B 1250 K
 C 500 K
 D 750 K
- Q 62 Saturated solution of KNO_3 is used to make salt bridge because
 A ☒ KNO_3 is highly soluble in water
 B Velocity of K^+ is greater than NO_3^-
 C Velocity of NO_3^- is greater than K^+
 D ☒ Velocity of both K^+ and NO_3^- are nearly same
- Q 63 If the rate of reaction between A and B is expressed as $K[A][B]^2$, then reaction is
 A First order in A
 B Second order in B ☒
 C Overall having third order
 D ☒ All are correct
- Q 64 Which of the following compound is amphoteric in nature
 A CH_3COCl
 B $(CH_3CO)_2O$
 C $CH_3COOC_2H_5$
 D ☒ CH_3CONH_2

- Q 65 Benzene diazonium chloride on reacting with phenol in weakly basic medium gives
A ☒ Diphenyl ether
B ☐ Benzene
C ☐ Chlorobenzene
D ☒ p-hydroxyazobenzene
- Q 66 Ammonia gas can be dried over
A ☐ CaCl_2
B ☐ Conc. H_2SO_4
C ☒ Quick lime
D ☒ PCl_5
- Q 67 The lanthanoid contraction is responsible for the fact that
A ☒ Zr and Y have about same radius
B ☐ Zr and Nb have similar oxidation state
C ☒ Zr and Hf have about same radius
D ☐ Zr and Zn have similar oxidation state
- Q 68 Which of the following has magnesium
A ☒ Chlorophyll
B ☐ Haemoglobin
C ☒ Vitamin B12
D ☐ Carbonic anhydrase
- Q 69 Which of the following is used as a control rod in nuclear reactor
A ☒ Boron
B ☐ Graphite
C ☐ Cadmium
D ☒ Boron + cadmium both
- Q 70 One of the following is a biodegradable polymer
A ☒ Cellulose
B ☐ Polystyrene
C ☐ Nylon-66
D ☒ PVC
- Q 71 Radial nodes present in 3s and 2p orbitals are respectively
A ☐ 0, 2
B ☒ 2, 0
C ☐ 2, 1
D ☐ 1, 2
- Q 72 Equal masses of methane and oxygen are mixed in an empty container at 25°C . The fraction of total pressure exerted by oxygen is
A ☒ $\frac{1}{3} \times \frac{273}{298}$
B ☒ $\frac{1}{3}$
C ☐ $\frac{1}{2}$
D ☐ $\frac{2}{3}$

$\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- Q 73 Anything influencing valence electrons will affect the chemistry of element. Which one of following factors does not affect the valence shell
A ☐ Valence principle quantum number(n)
B ☐ Nuclear charge(z)
C ☒ Nuclear mass
D ☐ Number of core electrons

- Q 74 For the reaction equilibrium $\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2 \text{NO}_2 (\text{g})$, the concentration of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol L}^{-1}$ respectively. The value of K_c for the reaction is
☒ A $3 \times 10^{-3} \text{ mol L}^{-1}$ ☐ B $3 \times 10^{-1} \text{ mol L}^{-1}$
☐ C $3 \times 10^{-2} \text{ mol L}^{-1}$ ☐ D $3 \times 10^{-4} \text{ mol L}^{-1}$
- Q 75 Which of the following is a redox reaction
☐ A $\text{NaCl} + \text{KNO}_3 \longrightarrow \text{NaNO}_3 + \text{KCl}$
☐ B $\text{CaC}_2\text{O}_4 + 2\text{HCl} \longrightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$
☐ C $\text{Ca}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \longrightarrow \text{CaCl}_2 + 2\text{NH}_3 + 2\text{H}_2\text{O}$
☒ D $2\text{K}[\text{Ag}(\text{CN})_2] + \text{Zn} \longrightarrow 2\text{Ag} + \text{K}_2[\text{Zn}(\text{CN})_4]$
- Q 76 In B_3H_6
☐ A $\text{H}_1 \dots \text{B} \dots \text{H}_6$ bond angle is 118°
☐ B Each boron atom undergoes sp^2 hybridisation
☒ C The boron atoms are linked through hydrogen bonds
☐ D All the atoms lie in the same plane
- Q 77 On strong heating sodium bicarbonate changes into
☐ A Sodium monoxide ☐ B Sodium hydroxide
☒ C Sodium carbonate ☐ D Sodium peroxide
- Q 78 The IUPAC name of $\text{C}_6\text{H}_5\text{COCl}$ is
☒ A Benzoyl chloride ☐ B Chlorophenyl ketone
☒ C Benzene carbonyl chloride ☐ D Phenyl chloroketone
- Q 79 Which of the following does not show geometrical isomerism
☐ A 1,2 - dichloro-1-pentene ☐ B 1,3 - dichloro-2-pentene
☒ C 1,1 - dichloro-1-pentene ☒ D 1,4 - dichloro-2-pentene
- Q 80 An organic compound having molecular mass 60 is found to contain C=20 %, H=6.67 % and N=46.67 %, while rest is oxygen. On heating it gives NH_3 along with a solid residue. The solid residue gives violet colour with alkaline copper sulphate solution. The compound is
☐ A $\text{CH}_3\text{CH}_2\text{CONH}_2$ ☒ B $(\text{NH}_2)_2\text{CO}$
☐ C CH_3CONH_2 ☒ D CH_3NCO
- Q 81 Formation of terylene is an example of
☒ A Condensation polymerisation ☐ B Addition polymerisation
☐ C Esterification ☒ D Saponification
- Q 82 Synthetic polymer that resembles natural rubber is
☒ A Neoprene ☒ B Buna-S
☐ C Glyptal ☐ D Nylon

- Q 83 Which law of thermodynamics introduces the concept of entropy
A First law ☐ B Second law ☒
C Third law ☐ D Zeroth law ☐
- Q 84 The standard emf for cell reaction $2\text{Cu}^+(\text{aq}) \longrightarrow \text{Cu}(\text{s}) + \text{Cu}^+(\text{aq})$ is 0.36 V at 298 K, the equilibrium constant of the reaction is
A 5×10^8 ☐ B 1.4×10^{12} ☐
C 7.4×10^{12} ☒ D 1.2×10^8 ☒
- Q 85 If the specific conductance and conductance of a solution is same, then its cell constant is equal to
A 1 ☒ B 0 ☐
C 1000 ☐ D 10 ☒
- Q 86 The rate equation for the reaction $2\text{A} + \text{B} \longrightarrow \text{C}$ is $\text{rate} = k[\text{A}][\text{B}]$. The correct statement in relation to this reaction is
A Unit of K must be s^{-1} ☐
B $1/2$ is constant ☒
C Rate of formation of C is twice the rate of disappearance of A ☐
D Value Of K is independent of initial concentration of A and B ☒
- Q 87 Which of the following undergoes mustard reaction
A $\text{C}_2\text{H}_5\text{NH}_2$ ☒ B $(\text{C}_2\text{H}_5)_2\text{NH}$ ☐
C $\text{C}_2\text{H}_5\text{CONH}_2$ ☐ D $(\text{C}_2\text{H}_5)_3\text{N}$ ☒
- Q 88 Hydrogen halide which cannot act as reducing agent
A HI ☐ B HBr ☐
C HF ☒ D HCl ☒
- Q 89 Which of the following radiations can be easily stopped by air
A α -rays ☒ B β -rays ☐
C γ -rays ☐ D All the above ☒
- Q 90 The atomic masses of elements are usually fractional because
A Elements consist of impurities ☐ B These are mixtures of allotropes ☐
C These are mixtures of isobars ☐ D These are mixtures of isotopes ☒

Biology

- Q 91 Resolving power of human eye is
☒ A 100 μm
☐ B 200 μm
☐ C 50 μm
☐ D 150 μm
- Q 92 Homeostasis is
☒ A Sweating during exercise
☐ B Heating of body during exercise
☐ C Intake of extra salt during summer and more fat during winter
☐ D All of above
- Q 93 Coliphage ϕ X 174 virus contains
☒ A Single stranded RNA
☐ B Double stranded RNA
☐ C Single stranded DNA
☐ D Double stranded
- Q 94 Steady state is represented by
☐ A Getting raw material
☐ B Intake of food and energy
☒ C Intake of material as well as energy and elimination of waste product
☐ D Removal of waste material and intake of raw material
- Q 95 9 + 2 arrangement is absent in
☐ A Flagellum of Chlamydomonas
☐ B Flagellum of Euglena
☒ C Flagellum of Paramecium
☐ D Flagellum of Bacterium
- Q 96 Immediate source of cellular energy is
☐ A Glucose
☐ B ATP
☐ C Pyruvic acid
☐ D NADH
- Q 97 The protein part of enzyme is
☐ A Prosthetic group
☒ B Apoenzyme
☐ C Holozyme
☐ D Zymogen
- Q 98 Respiratory substrate yielding maximum number of ATP molecules is
☐ A Ketogenic amino acids
☒ B Glucose
☐ C Amylose
☐ D Glycogen
- Q 99 Two kingdom classification was given by
☒ A Linnaeus
☐ B Haeckel
☐ C Whittaker
☐ D Lamarck
- Q 100 Heterocysts specialised for nitrogen fixation occur in certain
☐ A Red algae (*Bactrochasprium*)
☒ B Green algae (*Spirogyra*)
☐ C Blue green algae (*Anabena*)
☐ D Brown algae (*Laminaria*)
- Q 101 The infective stage of malaria parasite Plasmodium that enters human body is
☐ A Merozoite
☒ B Sporozoite
☐ C Trophozoite
☐ D Minata form

- Q 102 Sporangia bearing leaves in Pteridophytes are called as
 A Ramenta
☒ B Sori
 C Sporophyllus
 D Indusium
- Q 103 Largest amphibian is
 A Necturus
☒ B Icthyophis
 C Cryptobranchus
 D Amphiuma
- Q 104 Pyramid of energy is always
 A Inverted
 C Spindle shaped
☒ B Upright
 D Cup shaped
- Q 105 Symbiosis is observed between
 A Rat flea and rat
 B Sea anemone and hermit crab
☒ C Zoochlorella and Hydra
 D Leaf butterfly and the twig on which it rests
- Q 106 Red Data books are produced by
☒ A IUCN
 C IBWL
 B WWF
 D None of these
- Q 107 Vascular bundles in dicot root are
☒ A Radial exarch
 C Radial endarch
 B Conjoint
 D Conjoint exarch
- Q 108 ATP synthesis during light reaction is
☒ A Oxidative phosphorylation
 C Photophosphorylation
 B Photolysis
 D Phosphorylation
- Q 109 Number of calvin cycles required to generate a molecule of hexose is
☒ A 2
 C 6
 B 4
 D 8
- Q 110 C4 plants differ from C3 plants with respect to
 A First product
 C Number of ATP molecules consumed
 B Substrate which accepts carbon dioxide
☒ D All the above
- Q 111 Abscissic acid controls
 A Cell division
☒ C Leaf elongation
 B Leaf fall and dormancy
 D Root elongation
- Q 112 Haploid plant cultures are derived from
☒ A Leaves
 C Pollen grain
 B Root tip
 D Buds
- Q 113 The heart sound "DUPP" is produced by
☒ A Closure of semilunar valve
 C Closure of A-V valves
 B Opening of A-V valves
 D Opening of semilunar valve

- Q 114 Sarcolemma is a covering of
A Nerve fibre
C ☒ Muscle fibre
B Collagen fibre
D Spinal cord
- Q 115 Urine leaves kidney through
A ☒ Ureter
C Urethra
B Collecting duct
D Renal vein
- Q 116 In the mammalian brain the two cerebral hemisphere are joined by
A ☒ Corpus callosum
C Corpus quadrigeminum
B Corpus spongiosum
D Corpus cavernosum
- Q 117 According to immunity theory, ageing is due to
A Accumulation of errors
C ☒ Degeneration of thymus
B Reduced functioning of endocrine glands
D Accumulation of waste products
- Q 118 The correct sequence in the process of development is
A Cleavage-zygote- fertilisation-morula-blastula-gastrula
B ☒ Fertilisation-zygote-cleavage- morula-blastula-gastrula
C Fertilisation-cleavage-morula-zygote- blastula-gastrula
D Fertilisation-zygote- blastula- morula-cleavage-gastrula
- Q 119 Gene mutation occurs at the time of
A ☒ DNA replication
C DNA repair
B RNA transcription
D Cell division
- Q 120 Autocatalytic function is shown by
A ☒ Enzymes
C RNA
B DNA
D Hormones
- Q 121 The termination or stop codon in the genetic code is
A UAA
C UGA
B UAG
D ☒ All the above
- Q 122 Hugo de Veries worked on which plant
A ☒ *Oenothera*
C *Lathyrus*
B *Pisum*
D *Mirabilis*
- Q 123 Foetal abnormalities are caused by
A Nicotine
C ☒ LSD
B Alcohol
D Opium
- Q 124 Ectomycorrhizae are found in
A *Euclapytus*
C *Fagus*
B *Pinus*
D ☒ All the above
- Q 125 Cell organnele showing polymorphism is
A ☒ Lysosome
C Endoplasmic reticulum
B Golgi apparatus
D All of above

- Q 126 Bacteria lacking cell wall are
A Actinomycetes
C Chlamydiae
B Spirochaetes
D ☒ Mycoplasmas
- Q 127 Euchordata refers to
A All chordates
C Chordates and cephalochordates
B ☒ Vertebrates
D Hemichordates
- Q 128 The concept of ecological pyramids was given by
A Reiter
C Darwin
B Odum
D ☒ Elton
- Q 129 Cork cambium is
A Primary meristem
C ☒ Lateral meristem
B Ground meristem
D Intercalary meristem
- Q 130 Formation of fruits without fertilisation is called as
A ☒ Parthenocarpy
C Polygamy
B Parthenogenesis
D Polyembryony
- Q 131 Adrenalin gland is derived from
A Ectoderm
C ☒ Ectoderm and mesoderm
B Mesoderm
D Ectoderm and endoderm
- Q 132 Blood group agglutinin is a
A Phospholipid
C Haemoprotein
B Phosphoprotein
D ☒ Glycoprotein
- Q 133 Oxyntic cells secrete
A Trypsin
C NaOH
B ☒ HCl
D Pepsinogen
- Q 134 The number of sperms formed from four primary spermatocytes are
A 4
C ☒ 16
B 1
D 32
- Q 135 The unit of natural selection is
A Family
C ☒ Individual
B Specie
D Population

Mathematics

2005x 8/20

Q136 The value of $\cot^{-1} \left[\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}} \right]$ is

A $\pi - x$

B $2\pi - x$

C $\frac{x}{2}$

☒ D $\pi - \frac{1}{2}x$

Q137 If $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 \\ 1 & 0 \end{bmatrix}$, then

A $AB = BA = 0$

B $AB = 0, BA \neq 0$

☒ C $BA = 0, AB \neq 0$

D None of these

Q138 The principal value of $\cos^{-1} \left(-\frac{1}{\sqrt{2}} \right)$ is

☒ A $\frac{3\pi}{4}$

B $\frac{\pi}{4}$

C $-\frac{\pi}{4}$

D $-\frac{3\pi}{4}$

Q139 The roster form of a set $A = \{a_n : n \in \mathbb{N}, a_{n+1} = 3a_n \text{ and } a_1 = 2\}$ is

A $\{1, 2, 3, 4\}$

B $\{6, 18, 24, 30\}$

☒ C $\{2, 6, 18, 54, \dots\}$

D $\{1, 2, 3\}$

Q140 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Then :

A f is one-one onto

B f is many-one onto

☒ C f is one-one but not onto

D f is neither one-one nor onto

Q141 Let $A = \{1, 2, 3\}$, then the number of equivalence relations containing $(1, 2)$ is

A 1

☒ B 2

C 3

D 4

Q142 If $P(n)$ is the statement " $n(n+1)$ is even" then $P(3)$ is

A 6

☒ B $3(3+1)$ is even

C 5

D None of these

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2+2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + 2R_2$$

$$\begin{bmatrix} -2+1 & 0+0 \\ 0+0 & 0+0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$$

Q143 Principle argument of complex number $1 + \sqrt{-3}$ is

A $\frac{\pi}{6}$

B $\frac{-\pi}{4}$

C $\frac{\pi}{2}$

☒ D $\frac{\pi}{3}$

Q144 The roots of the quadratic equation $x^2 - 7ix - 12 = 0$, are

A $-3i, 4i$

☒ B $3i, 4i$

C $2i, 3i$

D None of these

Q145 The sum of all natural numbers between 1 and 99 which are multiples of 5 (or divisible by 5) is

A 75

B 365

B 950

D 100

Q146 If the straight line $y = mx + c$ passes through the points (2,4) and (5,0), then the value of m and c are

A $m = 3, c = \frac{3}{5}$

B $m = -2, c = 0$

C $m = 0, c = -4$

☒ D $m = \frac{-2}{5}, c = \frac{24}{5}$

Q147 The equation of parabola that satisfies the condition, focus (0,-3) and directrix $y=3$, is

A $y^2 = -12x$

B $x^2 = 12y$

☒ C $x^2 = -12y$

D $y^2 = 6x$

Q148 The co-ordinates of the foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

A $(0, \pm ae)$

B $(\pm a, 0)$

C $(0, \pm a)$

☒ D $(\pm ae, 0)$

Q149 Equation of circle of radius r touching both axes is

A $x^2 + y^2 - 2hx - 2ky = 0$

☒ B $x^2 + y^2 - 2rx - 2ry + r^2 = 0$

C $x^2 + y^2 - 2ax = 0$

D None of these



150 For the function $f(x) = x \cos \frac{1}{x}$, $x \geq 1$ $f'(x)$ is

- A Strictly increasing in $(1, \infty)$
 C Strictly increasing in $(0, \infty)$

- ☒ B Strictly decreasing in $(1, \infty)$
 D None of these

$$f(x) = x \cos \frac{1}{x}, x \geq 1$$

$$f'(x) = -\sin \frac{1}{x}$$

151 The integrating factor of $x \frac{dy}{dx} + 2y = x^2$

- A x
 C e^x

- ☒ B e^{-x}
 D x^2

$$x \frac{dy}{dx} + 2y = x^2$$

5) 52 Every continuous function is

- A Increasing
☒ C Not differentiable always

- B Decreasing
 D Differentiable

53 If $y = \cos^{-1}(4x^2 - 3x)$, then $\frac{dy}{dx}$ is equal to

- A $3/\sqrt{1-x^2}$
 C $\sqrt{1-x^2}$

- ☒ B $-3/\sqrt{1-x^2}$
 D None of these

$$x \int \frac{dy}{dx} + 2y = x^2$$

$$x^2/2 + y = x^3/3$$

$$y = \frac{x^3}{3} - \frac{x^2}{2}$$

54 If $|\vec{a}| = 3$ and $|\vec{b}| = \frac{\sqrt{2}}{3}$, then $\vec{a} \times \vec{b}$ is a unit vector, if angle between $\vec{a}$ and $\vec{b}$ is unit vector if

- A $\frac{\pi}{6}$

- B $\frac{\pi}{3}$

- ☒ C $\frac{\pi}{4}$

- D $\frac{\pi}{2}$

55 If $\vec{a}$ is a non-zero vector of magnitude 'a' and λ a non-zero scalar, then $\lambda \vec{a}$ is a unit vector if

- ☒ A $a = \frac{1}{|\lambda|}$

- B $\lambda = 1$

- C $\lambda = -1$

- D $a = |\lambda|$

generations of a family is called the **pedigree analysis**. In the pedigree analysis the inheritance of a particular trait is represented in the family tree over generations.

In human genetics, pedigree study provides a strong tool, which is utilised to trace the inheritance of a specific trait, abnormality or disease. Some of the important standard symbols used in the pedigree analysis have been shown in Figure 5.13.

house studied in this chapter, each and every feature in any

19

SET-A

UET-2018(UG-Programme)

- Q156 The distance of the point $(2, 3, -5)$ from the plane $x + 2y - 2z = 9$, is
- A $\frac{13}{3}$ Units ☒ B 3 Units
- C 2 Units D None of these
- Q157 The planes $2x - y + 4z = 5$, and $5x - 2.5y + 10z = 6$ are
- A Perpendicular ☒ B Parallel
- C Intersect y-axis D None of these
- Q158 Objective function of a L.P.P is
- A A Constraint ☒ B A function to be optimized
- C A relation between variables D None of these
- Q159 The solution set of the inequation $x + 2y > 3$ is
- A Half plane containing the origin B First quadrant
- ☒ C Open half plane not containing the origin D None of these
- Q160 The minimum value of $z = 2x + 3y$ subject to the constraints $1 \leq x + 2y \leq 10, x \geq 0, y \geq 0$
- A 20 ☒ B 1.5
- C 5.1 D 12
- Q161 The linear inequalities on the variable x and y are known as
- A Feasible Region B In feasible region
- ☒ C Constraints D None of these
- Q162 Which axes is contained by half plane representing the inequation $y - x \leq 0$
- ☒ A Positive x-axis B y-axis
- C x-axis D Positive y-axis
- Q163 In the given statement or is an inclusive or exclusive "Two lines intersect at a point or are parallel"
- A Inclusive B Both
- ☒ C Exclusive D None of these
- Q164 In the given statement "There exists" is a, "There exists a number which is equal to its square"
- ☒ A Quantifier B Connectives
- C Not Connective D None of these
- Q165 If two or more simple statements are combined by the use of words such as "and", "or", "not", "if", "then", "if and only if", then the resulting statement is called
- A Connective statement ☒ B Compound statement
- C Negation statement D New Statement

below which is the epidermis, two muscle layers (circular and longitudinal) and an innermost coelomic epithelium. The epidermis is made

- 166 The inverse of the statement: "if $x < y$, then $x^2 < y^2$ "
 A if $x^2 < y^2$, then $x < y$ B if $x < y$, then $x^2 > y^2$
 C if $x < y$, then $x^2 < y^2$ D if $x^2 < y^2$, then $x < y$
- 167 The marks obtained by 20 students in a test were
 13, 17, 11, 5, 18, 16, 11, 14, 13, 12, 18, 11, 9, 6, 8, 17, 21, 22, 7, 6 then mean marks equal to
 A 31 B 25
 C 12.04 D 12.75
- 168 The median of the given data is
 15, 5, 7, 3, 11, 28, 9, 43, 28, 17, 31
 A 15 B 43
 C 28 D 11

- 169 If A and B are events such that $P\left(\frac{A}{B}\right) = P\left(\frac{B}{A}\right)$, then
 A $A \subset B$ but $A \neq B$ B $A = B$
 C $A \cap B = \emptyset$ D $P(A) = P(B)$
- 170 Three coins are tossed once, probability of getting exactly 2 heads is
 A $\frac{3}{8}$ B $\frac{1}{2}$
 C $\frac{1}{8}$ D $\frac{1}{4}$

- 71 If $P(A) = \frac{6}{11}$, $P(B) = \frac{5}{11}$, $P(A \cup B) = \frac{7}{11}$ then $P\left(\frac{B}{A}\right)$ equals to
 A $\frac{5}{11}$ B $\frac{2}{5}$
 C $\frac{3}{4}$ D $\frac{2}{3}$

(A) $\frac{6}{11}$, $P(B) = \frac{5}{11}$
 (TIT) (HHH)
 (THT) (HTH)
 (TTH) (HHT)
 (HTT) (THH)

- 72 If two forces of magnitude 4 and 5 have a resultant of magnitude 6, then the angle between the two forces is
 A $\pi - \cos^{-1}\left(-\frac{1}{8}\right)$ B $\pi - \cos^{-1}\left(\frac{1}{8}\right)$
 C $\cos^{-1}\left(\frac{1}{2}\right)$ D None of these

- 3 The moment about a point O of a force of 20 newton acting at a point along a line AB when $OA = 3$ m and $\angle OAB = 90^\circ$, is
 A 30 newton-metre B 12.2 newton-metre
 C 16 newton-metre D 60 newton-meter

- Q174 Two forces acting at point A(2,0) and B(-2,0) form a couple of moment 20 units, if $|AB|$ is the arm of the couple, then forces are
 A 12 Units B 4 Units
 C 5 Units D None of these
- Q175 P, Q are the magnitude of two unlike parallel forces. When P is doubled, it is found that the line of action of Q is midway between the lines of actions of the old and new resultant. Then the ratio of P to Q is
 A 3:4 B 2:3
 C 5:4 D 3:7
- Q176 If a particle is moving in a straight line and its distance from a fixed point O on the line after t seconds is given by $s = 2 - 3t + 4t^2$, then its acceleration at the end of 2 seconds is
 A 32 cm/sec² B 22 cm/sec²
 C 48 cm/sec² D 24 cm/sec²
- Q177 A body is thrown vertically upward with a velocity of 24.5 m/sec, then the maximum height reached is
 A 30.52 m B 30.165 m
 C 30.625 m D 30.47 m
- Q178 A stone is thrown vertically upwards with a velocity 98 m/sec, then the interval of time between the two instants when it is 313.6 m from the ground
 A 6 sec. B 3 sec.
 C 4 sec. D 12 sec.
- Q179 A thief jumps off the terrace of a building with a heavy suitcase and falls freely under gravity while he is falling, then pressure on his head is
 A Equal to the weight of suitcase B 9.6 kg wt.
 C No Pressure D None of these
- Q180 A stone dropped from a balloon moving horizontally with a velocity of 40 m/sec reaches the ground in 5 sec., then the height of the balloon is if $g = 10 \text{ m/sec}^2$
 A 12.5 m B 1.25 m
 C 125 m D None of these

